Cusum - Entropy: An efficient method for DDoS attack detection

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***Abstract*—Computers and Internet have evolved into necessary tools for our professional, personal and social lives. As a result of growing dependence, the availability of these systems has become a concern. This concern increases exponentially when considering systems such as smart power grids and public utility services. Therefore, research should be conducted to develop effective ways of detecting system anomalies. Denial of Service attacks are an important problem for the communication systems. Researchers proposed many DDoS detection approaches. In this study, we propose a novel DDoS detection approach: Cusum - Entropy. Our approach performs additional signal processing on the entropy of the packet header field to improve detection efficiency. We tested our approach using operational network traffic and performing DDoS attacks without jeopardizing the operation network. Our results showed that our approach gives high detection and low false positive rates and outperforms the detection approach using the entropy of packet header field.**

***Index Terms*--Change detection algorithms, Communication system security, Entropy, Industrial communication, Software defined networking**

I. INTRODUCTION

Security incidents on industrial and energy systems, such as Stuxnet [1] and North East US blackout [2] show effective security methods are necessary to protect Supervisory Control and Data Acquisition (SCADA) systems. Denial of Service attacks are an important problem for the communication systems. A Distributed Denial of Service (DDoS) attack disables network services for legitimate users. While DDoS attacks cause inconvenience for users, and revenue loss for service providers; their effects on critical infrastructures like smart grid and public utilities could be catastrophic. For example, an attack on a smart grid system communication can cause cascaded power failures and lead to major blackout.

Freely available Denial of Service (DoS) attack tools like Stacheldraht [3] and Low Orbit Ion Cannon (LOIC) [4] make it possible for unsophisticated users to perform these attacks. According to Arbor Networks 2011 Worldwide Infrastructure Security Report, the number of flooding DDoS attacks in the 10 Gbps range have increase significantly [5] and the highest bandwidth observed for a single attack has reached 300 Gbps [6]. Today, existing DDoS attacks are capable of throwing small countries off-line [7] and these attacks are considered a major threat to governments and large corporations. Political and ideological differences, inter-personal/group conflicts and extortion are common motivations for these attacks. DDoS attacks against Estonia (2007) and Georgia (2008) showed the importance of the DDoS threat at the national level.

These attacks can be performed by altering the configuration files of compromised nodes, physically damaging network components or consuming resources [8]. Carl et al., divided resource starvation attacks into two general categories; vulnerability attacks and flooding attacks [9]. Vulnerability attacks leverage software or protocol bugs to exhaust system resources, such as memory, CPU time, disk space or data structures. Flooding attacks send more packets or requests than the system can handle.

If the attacker uses multiple nodes to perform a DoS attack, it is called Distributed Denial of Service (DDoS) attack. Nodes compromised by the attacker become zombie agents in a bot-net. To perform the attack, zombies send dummy traffic/requests to the victim at the attacker's command.

In this paper we present a novel DDoS detection approach cusum - entropy. We used Cumulative Sum (CUSUM) algorithm [10] on entropy of the observed network traffic after a pre-filtering stage using wavelet filter. Our results showed that the cusum - entropy approach detects DDoS attacks with high detection efficiency. Also, wavelet filtering and cusum steps improve the detection performance and gives better results than the detection approaches using network traffic entropy without additional signal processing.

The rest of the paper is organized as follows: Section 2 gives a review of recent DoS detection studies. Section 3 explains the detection algorithm proposed in this study and the background on entropy, wavelet and cusum. Section 4 describes the experiment setup, testing procedure and presents the results. Finally, the paper is concluded in the Section 5.

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II. RELATED WORK

DoS detection can be put into two different categories: signature detection and anomaly detection. Signature detection compares network traffic with the known attack templates to detect attacks. In anomaly detection, the detector uses network traffic statistical features; such as entropy of incoming packet source IP addresses. It continuously calculates the observed statistical feature and sends an alarm if an extreme deviation occurs. While in most cases anomaly detectors have a higher false positive rate than signature detectors, they can detect attacks previously unknown by the system.

Many entropy based anomaly detection approaches have been proposed in the past decade [11]-[16]. Lee and Xiang investigated some information theoretic measures; entropy, conditional entropy, information gain, information cost; and their use cases for anomaly detection [11]. Nychis et al. analyze the detection capabilities and correlations of different entropy based metrics, such as different packet header fields [12].

Oshima et al. used short-term entropy for early detection of DDoS attacks [13]. Authors classified the attacks using distributions of the average and standard deviation of the entropy. In [14], they also proposed a dynamic threshold calculation method for short-term entropy based DDoS detection method.

Prasad et al. presented an entropy based DDoS attack detection and trace back approach using Threat Monitor System [15]. Tritilanunt et al. minimized the false positive rate of the IPSE-based approach by additionally using packet content observations [16].

III. CUSUM - ENTROPY

In this section we present our novel DDoS detection approach: Cusum - Entropy and the background information about entropy, wavelet and CUSUM.

A. Entropy

Entropy was defined by Shannon in 1948 [17] as a measure of uncertainty. Let X be a dataset with a finite number of independent symbols from 1 to n, $\{x_1, x_2 \dots x_n\}$, and the probability of these symbols are $p = \{p_1, p_2 \dots p_n\}$ respectively. Entropy of X is:

$$H(X) = -\sum_{i=1}^{n} p_i \log p_i \qquad (1)$$

The entropy value is largest when X has a uniform distribution. It becomes 0 if one symbol has probability 1. For all other distributions of X, entropy varies between 0 and the maximum entropy value. The entropy value found using "(1)" ranges between 0 and $\log_n$. To make entropy values independent of the number of distinct symbols, the entropy can be normalized to vary from 0 to 1:

$$H_N(X) = \frac{H(X)}{\log n} \qquad (2)$$

where n is the number of distinct symbols in an observation window.

In DDoS detection, entropy measures the amount of disorder in the observed data. However, even if the symbols have a uniform distribution, a different number of unique symbols in an observation window will force "(1)" to compute a different entropy value. Therefore, in our study we used normalized entropy.

Entropies of packet header fields; such as source/destination IP address, and protocol type change significantly during DDoS attacks. The entropy change during an attack varies based on observed packet header field. While the entropy of the source IP addresses increases during a DDoS attack, it causes destination IP address entropy to decrease. In this study, we used entropy of source IP address but our method can be used with entropy of other packet header fields; such as source port and protocol type.



Figure 1: Cusum – Entropy method flow diagram

B. Wavelet

The wavelet transform analyzes an input signal simultaneously in the time and frequency domains. It differs from the Fourier transform by localizing the frequency components of the signal in the time.

Haar wavelet coefficients are calculated by averaging and differencing two data values. Averaging gives the low-pass, and differencing provides high-pass signal characteristics [18]. Also, the coefficient obtained by differencing represents the average amount of signal change between two sample if Wavelet Decomposition Level (WDL) is 1 or two averages, if WDL > 1.

As an example, Haar wavelet decomposition of the data set $\{x_1, x_2\}$ is the combination of the average and the difference of the x_1 and x_2 and the final result is $\left\{\frac{x_2+x_1}{2}, \frac{x_2-x_1}{2}\right\}$

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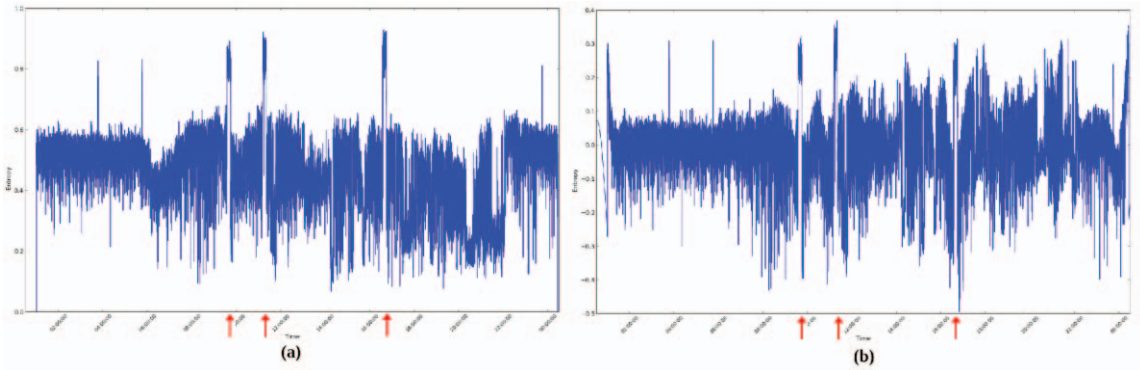


Figure 2: Wavelet filter is used to filter out long term trends of entropy data, which leads to a very poor performance of CUSUM algorithm. An Entropy time-series before and after the wavelet filter block. In the figures arrows show the attack times. (a) An Entropy time-series with three DDoS attacks before the wavelet filter. (b)An Entropy time-series with three DDoS attacks after the wavelet filter.

In our study we used wavelet to filter out the long term variations of the observed entropy values to reduce the number of false alarms. Then the filtered signal is fed into a cusum algorithm for detection.

C. Cumulative Sum (CUSUM)

The cusum algorithm is used to detect extreme increases hidden in bursty background data. In our study we used the cumulative sum (CUSUM) approach proposed by Blazek et al. [10]. Their method calculates the difference between the current and long-term average of the observations. If the current average increases faster than long-term average, the cusum coefficient also increases. The cusum coefficient goes back to zero when the difference between two averages is small. When the cusum coefficient exceeds the chosen threshold, it indicates an entropy increase which may be caused by a DDoS attack. This cusum process can be summarized as:

$$S[t] = \max\{0, (S[t-1] + H[t] - m[t])\};\ S(0) = 0 \qquad (3)$$

where S[t−1] is the old cusum value, H[t] is the entropy value at time t and m[t] is the long-term average of CUSUM input, calculated with a given long term averaging memory (ε) between $0 < \varepsilon < 1$;

$$m[t] = \varepsilon\, m[t-1] + (1-\varepsilon)H[t];\ m[0] = 0 \qquad (4)$$

To reduce high frequency noise H[t], is low-pass filtered using local averaging memory (α), $0 < \alpha < 1$:

$$\tilde{H}[t] = \alpha\, H[t] + (1-\alpha)\tilde{H}[t-1];\ \tilde{H}[0] = 0 \qquad (5)$$

High values for ε make the average emphasize the long term average value in m[t]. Larger values for α make the average emphasize the current value in H[t].

Substituting "(5)" for H[t] and adding algorithm correction variable C in "(3)" gives us the modified Cusum algorithm:

$$\tilde{S}[t] = \max\{0, (\tilde{S}[t-1] + \tilde{H}[t] - m[t] - C)\};\ \tilde{S}(0) = 0 \qquad (6)$$

where C is multiplication of m[t] and correction parameter (ce) which forces the cusum coefficient values to 0 by adding more weight to long term average, (m[t]).

D. Cusum – Entropy Algorithm

Cusum - Entropy detection method flow diagram is in Fig. 1. The entropy module calculates the entropy of a packet header field from the packets in an observation window. In our study we used the entropy of source IP address. To calculate entropy of the source IP address; unique source IP addresses and number of occurrence of these IP addresses in the observation window are determined. The probability of observing a source IP address in the observation window is calculated by dividing the number of times the source IP address was observed by the total number of observations. After calculating all unique source IP address probabilities, we calculated the entropy of the observation window using "(1)" and normalized entropy using "(2)".

Wavelet filter is used to filter out long term trends of entropy data, which leads to a very poor performance of CUSUM algorithm. The entropy data is decomposed into its high-pass and low-pass components as explained in Section 3.A. In our study, we performed ten step wavelet decomposition and filtered the tenth level low-pass components. Then the signal was reconstructed to get filtered entropy data. An entropy time-series with three DDoS attack before and after the wavelet filter is presented in Fig. 2.

Finally, filtered entropy data is processed using the Cusum algorithm. We set the cusum long term averaging memory (ε) and cusum local averaging memory (α) values 0.9 to see the extreme entropy deviations. Also, we used cusum correction

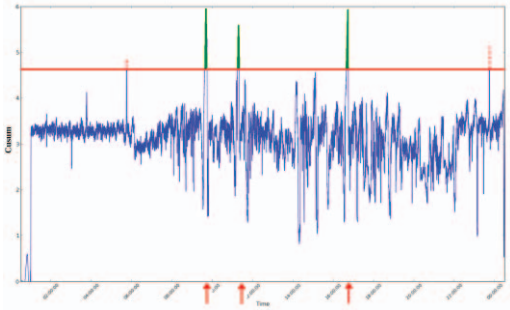


Figure 3: A cusum time-series with three DDoS attacks. In the figures the horizontal line is the optimal threshold. True detections are presented with solid line and false positives are shown with dashed line above the threshold.

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parameter (ce) = 0.1 to keep cusum coefficient values at 0 when long term and local average values of entropy data close to each other. Calculated cusum coefficients are used to detect DDoS attack. A cusum time-series with three DDoS attack is presented in Fig. 3. Attacks are shown using arrows and the optimal detection threshold is the horizontal line. The value above the threshold is a plotted dashed line if it is a false positive or solid line if it is a true detection.

IV. TESTING AND RESULTS

In anomaly based DDoS detection, detectors calculate the deviation of the observed statistical features from background traffic statistics to infer anomalies. It is essential to use operational background traffic for developing and testing these detection methods. Therefore, we tested the performance of our detection approach using operational background traffic and performing DDoS attacks without jeopardizing the original network. We observed the entropy of source IP address in non-overlapping time windows. These observations are saved with timestamps to generate source IP address entropy time-series.

A. Test Setup

The test setup is in Fig. 4. All switches are controlled by one Openflow [19] controller. The controller-switch logical connections are shown with dotted lines. In the figure, Clemson University network traffic and Clemson Condor cluster represent background and attack traffic sources respectively.

B. Background Traffic

A mirrored copy of Clemson University campus network traffic was used as background traffic during the tests. The amount of traffic generated in a network depends on the allowed network services (e.g. email, cloud services, VoIP, downloading/uploading large files), and the average number of users. We used 30 Mbps operational network traffic. This is typical amount of background traffic on building links at Clemson University. We believe the amount of background traffic used is typical for large companies/institutions.

C. Attack Traffic

The attack traffic is generated using the Condor [20] cluster to flood the link between switches. The Condor computer cluster has a hierarchical structure. Its work flow is very similar to a botnet. Namely, the user sends his request to the condor submit node with all executable and configuration files. This node assigns task(s) to its proper agent(s) based on the configuration file given by user. During our tests, we used up to 200 distinct condor nodes to perform UDP flooding attacks and generated average 416 Mbps attack traffic.

D. Testing Procedure

In this study, 12 source IP entropy time-series containing 33 DDoS attack were used. All time-series are one day long and sampling frequencies are two seconds. For analysis, attack start/stop time and attacker node count information was recorded for each attack.

The false positive and the true positive rates are found by comparing the calculated coefficient with given threshold. If the coefficient is bigger than the threshold, we decide if it is a true or a false detection. A true detection is decided when the coefficient time stamp is in the recorded attack time; it is decided as a false detection otherwise. Outside the recorded attack regions, time-series are split into small time intervals (I_{nr}).

Only the first true and the first false detection is counted in a recorded attack time and Inr respectively. True positive (TPR) and False positive (FPR) rates are:

$$\text{TPR} = \frac{\textit{Number of true detections}}{\textit{Number of attacks}} \tag{7}$$

$$\text{FPR} = \frac{\textit{Number of false detections}}{\textit{Number of nonattack time intervals } (I_{nr})} \tag{8}$$

In Equation 10, calculated FPR value changes with the I_{nr} size change. Small I_{nr} sizes increase total number of I_{nr} in non-attack region and results in artificially small FPRs. On the other hand, increase in I_{nr} size leads to higher FPR values and information loss by counting multiple false alarms as one. To decide I_{nr} size, we investigated false positive change rate depending on the I_{nr} size. While the false positive change rate is big at small I_{nr} sizes, it gradually become smaller as I_{nr} size gets bigger. In our study, we used the I_{nr} size 1800 seconds where false positive change rate starts to slow down.

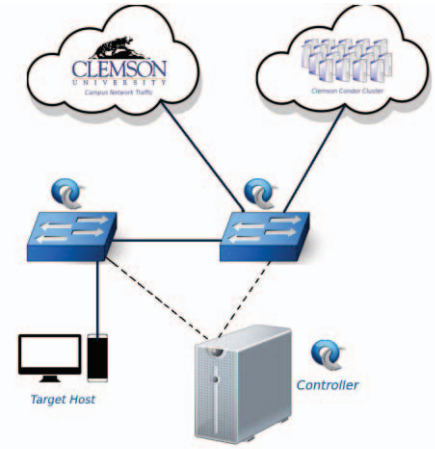


Figure 4: Test setup and controller logical connection

E. Testing Results

Our results are presented using Receiver Operating Characteristic (ROC). ROC curves present the False and True positive rate variations with threshold change. False and True positive rates vary between zero and one. The best performance is observed when TPR is 1 and FPR is 0; (1,0) point. Smaller minimum distance between an ROC curve and point (1,0) infers better performance. We compared performance efficiency of our DDoS detection approach with a detection approach using entropy of source IP address without further processing. Fig. 5 presents the performance of both detection approaches. Our detection approach gives higher detection and lower false positive rate.

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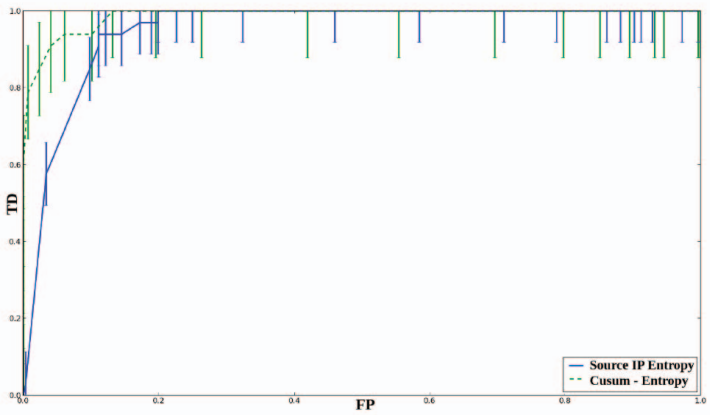


Figure 5: Detection efficiency of cusum - entropy approach and detection approach using entropy of source IP address with 95% confidence. Solid line: detection approach using entropy of source IP address. Dashed line: cusum - entropy approach.

V. CONCLUSION

Computers and networks are crucial part of the control and coordination part of critical infrastructures. As a result of this growing dependence, we need a better understanding of these systems and their reactions under unexpected conditions. Therefore there is a definite need for research and development of methods to analyze and protect these systems. According to the 2013 Prolexic Quarterly Global DDoS Attack report, the total number of attacks rose 21.75% compared to same quarter in 2012 [21]. Researchers proposed many DDoS detection approaches to address this issue. Entropy based detection is one of the common approaches studied in the past decade. In this study, we proposed a novel DDoS detection method: Cusum - Entropy. The additional signal processing we applied on observed entropy data improves detection efficiency. Our results show that Cusum - Entropy approach detects attacks with high detection and low false positive rates. Also Cusum Entropy approach gives better detection efficiency than a detection approach using entropy of packet header field without further processing. We used entropy of source IP address in this study but this approach can be generalized to entropy of other packet header fields.

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